

MLJ Contribution Information Sheet

Manuscript: Causal Transformer: Scaling Gradient-Based
Causal Discovery to 500 Variables

Corresponding Author: Shuaidong Gao

1. What is the main contribution of this paper and why is it important?

The main contribution is the introduction of Causal Transformer (CT), a self-attention architecture that treats variables as tokens and learns causal edge strengths from multi-head attention weights. CT retains NOTEARS’ acyclicity constraint $h(W)$ but constructs the adjacency matrix W through attention rather than independent parameter optimization, providing d^2 pairwise gradient pathways per edge. This matters because NOTEARS—the most widely used differentiable causal discovery method—collapses to exactly zero edges at $d \geq 150$, leaving $d = 200$ –500 inaccessible to DAG-based methods. This range is where most real transcriptomic and neuroimaging datasets sit. CT is the first method to operate reliably in this regime, validated by 335 experiments on synthetic, TCGA, and multi-omics data.

The contribution is supported by seven lines of evidence: (i) an 80-configuration synthetic data sweep (5 seeds per configuration) establishing the sharp viability boundary at $d = 200$, (ii) multi-seed TCGA-BRCA scaling at $d = 250$ –350 (5 seeds each) confirming CT’s unique operability where NOTEARS collapses, (iii) multi-omics validation (expression, CNV, methylation) revealing modality-specific limits, (iv) MLP baseline confirming attention-specific advantages over comparable-capacity alternatives, (v) validation on 10 TCGA cancer types with 100% positive gain over NOTEARS, (vi) external validation of top hub genes against ClinGen (75%), DrugBank (68%), and COSMIC (86%), and (vii) comprehensive ablation (5 studies) and convergence analysis (97.5% DAG-valid).

2. What evidence supports this contribution?

The evidence is organized into seven tiers. First, an 80-configuration sweep ($d \in \{30, 50, 100, 200\}$, $n \in \{200, 500, 1000\}$, 10 attention strategies, 5 seeds at all d) establishes the viability boundary. The best strategy (d128_h8_e500) recovers 1,028 edges at $d = 200$, $n = 1000$ versus NOTEARS’ 0. Second, TCGA-BRCA scaling at $d = 250$ –350 (5 seeds each) confirms CT’s unique operability. Third, BRCA multi-omics validation (expression, CNV, methylation; 5 seeds each) identifies a modality-specific failure mode. Fourth, an MLP baseline confirms that CT’s advantages are architectural, not parametric. Fifth, 10 TCGA cancer types at $d = 200$ show 100% positive gain (mean 198 edges vs. NOTEARS 0), with cluster coherence of 0.73. Sixth, top hub gene validation shows $3.5\times$ enrichment over random gene sets ($p < 10^{-4}$). Seventh, 97.5% of configurations converge to DAG-valid solutions across 335 total experiments.

3. What are the most closely related prior papers, and what is the exact relationship?

The most directly related prior works are:

1. Zheng et al. (2018) “DAGs with NO TEARS” (NeurIPS)—the foundational framework whose $O(d^3)$ matrix exponential bottleneck CT replaces. CT uses NOTEARS’ augmented Lagrangian for DAG enforcement but substitutes self-attention for the edge-weight optimization.
2. Gao (2026) “Cluster-Aware Gating” (CAGate)—embedding sample clustering into NOTEARS. CAGate addresses $d < 200$ through cluster-aware loss weighting; CT addresses $d = 200$ –500 through a fundamentally different computation graph.
3. Gao (2026) “SSCAGate”—extends CAGate with a sample-size phase transition theory. SSCAGate provides theoretical guarantees at $d < 200$; CT operates in the regime where SSCAGate’s NOTEARS engine collapses.

4. Vaswani et al. (2017) “Attention Is All You Need” (NeurIPS)—the original Transformer architecture. CT adapts multi-head self-attention to causal discovery by treating variables as tokens without positional encoding.

This submission has not appeared previously as a conference paper. No part of this work is currently under review elsewhere.